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## **RESEARCH ARTICLE**

# **Soxhlet-Assisted Matrix Solid Phase Dispersion for the quantitative analysis of 2-Ethylhexan-1-ol**

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### **ABSTRACT:**

**Background/Objectives:** 2-Ethylhexan-1-ol (2-EHN) is a fatty alcohol, an organic compound used as a frother during the production of ground calcium carbonate (GCC) to remove impurities.

**Methods/Statistical analysis:** Matrix solid-phase dispersion (MSPD) using XAD-4 (a non-polar polystyrene resin) was applied to adsorb and concentrate 2-EHN followed by Soxhlet extraction that extract 2-EHN (having a limited solubility for water) by the continuous extraction with recycling the solvent (ethyl acetate). The mass spectral ions of 2-EHN (acquired by GC-MS), m/z 57, 70, 83, 98, and 112 were accessed for the most accurate quantification.

**Findings:** The amount of 2-EHN adsorbed to XAD-4 resin (1 g) was highest after stirring for 14 hours followed by continuous extraction with 80 mL of ethyl acetate for 3 hours at 90°C. A modified Soxhlet extractor with a glass thimble (having a small hole at the bottom) worked well. The most accurate results were given by the ions at m/z 83 and 98 with 126% each (the value closest to 100%). These may be the most desirable mass spectral ions for the quantitation of 2-EHN because of the lack of background noise. The calibration function was linear with a correlation coefficient of 0.9944. The limit of quantitation was measured as 0.5 µg/L. The established 2-EHN analysis method was successfully applied to real surface water sampled far from and close to a ground calcium carbonate (GCC) quarry, where the resulting levels of 2-EHN were shown to be 3.0 and 6.8 µg/L, respectively.

**Improvements/Applications:** 2-EHN in environmental surface water was accurately quantified by GC-MS analysis using m/z 83 or m/z 98 after sample treatment by adsorption to Amberlite XAD-4 and continuous Soxhlet extraction with ethyl acetate enabling ppb level quantitation.

**KEYWORDS:** MSPD, Soxhlet extraction, 2-Ethylhexan-1-ol, GC-MS, Surface water.

### **1. INTRODUCTION:**

Liquid-liquid extraction (LLE) is a simple method (usually carried out by manual shaking with a separatory funnel) to extract by partition target compounds from a liquid phase, in which they are weakly dissolved, to an immiscible liquid phase having better solubility of the target substances.

However, LLE has some imperfections such as emulsion formation and insufficient removal of unwanted matrix material. Matrix solid phase dispersion (MSPD) is a powerful technique for disrupting and extraction of solid, semi-solid and viscous samples<sup>1</sup>. MSPD was introduced in 1989 and it is the one of common sample preparation technique due to its simplicity and flexibility over more classical method for these purposes. According to Baker<sup>2</sup>, "MSPD is based on several simple principles of chemistry and physics, involving forces applied to the sample by mechanical blending to produce complete

sample disruption and the interactions of the sample matrix with a solid support bonded-phase (SPE) or the surface chemistry of other solid support materials". MSPD is preferably used because of its easiness, selectivity, and a combined step for simultaneous extraction and clean-up, which results in rapid pre-treatment and low solvent consumption<sup>3</sup>. During 1999 Belgian dioxin crisis (mg/kg levels of PCBs were contaminated in animal feed, chicken fat, and eggs), Pat Sandra developed MSPD using acidic silica gel (44% sulfuric acid) for the high-throughput analysis of polychlorinated biphenyls (PCBs) 28, 52, 101, 118, 138, 153 and 180 to estimate the dioxin contamination level<sup>4</sup>. About 4000 suspected samples to be contaminated with dioxins were analyzed by Sandra and David<sup>3</sup> for PCBs by low resolution gas chromatography – mass spectrometry (GC-MS) analysis after MSPD sample preparation with low cost which is tens of times cheaper than dioxin analysis by GC-high resolution MS after very long and tedious sample preparation steps.

The benefits of MSPD, that it is quick and easy to handle with a low cost, are well proven, and the MSPD technique has been utilized as "QuEChERS" (quick, easy, cheap, effective, rugged, and safe). This is a famous pesticide sample preparation method about which Anastassiades et al.<sup>5</sup> claimed, "We feel that we have successfully fulfilled our aim to develop a method that is rapid, simple, inexpensive, effective, safe, potentially rugged, uses minimal amounts of solvents, needs no special equipment, avoids glassware (and cleaning/storage thereof), and still provides high quality results for a wide range of pesticides in foods". However, the recovery of the target substance(s) from the solid sorbent (used in MSPD) is not always easy and quick depending on the chemical and physical characteristics of the target compounds and solid sorbents.

Especially the target compounds, tightly bounded onto (or into) solid sorbents, should be recovered by very vigorous methods such as Soxhlet extraction and accelerated solvent extraction. Soxhlet extraction is named for its inventor, Franz Ritter von Soxhlet (1848–1926), a German chemist who worked on issues of milk chemistry. He developed the procedure, first described in 1879, as a way to separate the fats from milk solids. It remains a popular and important method for chemistry, biochemistry, and industry, particularly as they relate to food, plastic, and oil<sup>6</sup>. It is an efficient way to extract target compounds having a limited solubility for the solvent with continuous extraction by recycling the solvent. It is a continuous extraction method using refluxed solvent extraction for the solid sample within the heated solvent reservoir, a cold condenser, and a thimble containing the sample.

The fatty alcohol 2-ethylhexanol (2-EHN; CAS No. 104-76-7) is an organic compound used in the manufacture of a variety of products. The primary use of 2-EHN is in the manufacture of the diester bis (2-ethylhexyl) phthalate (DEHP), a plasticizer<sup>7</sup>. Additionally it can be also used as a frother during the production of ground calcium carbonate (GCC) for the removal of impurities. Froth flotation is an important concentration process that selectively separates hydrophobic valuable minerals from hydrophilic waste gangue. In its simplest form, froth flotation is a method where minerals can be "skimmed" from the surface of "slurry" that is "foamed" with the assistance of specific chemicals, water, and air bubbles<sup>8</sup>.

Therefore there is a possibility that contamination of surface water with 2-EHN could give rise to an increase in turbidity. The standard method for monitoring 2-EHN in air employs the use of charcoal solid sorbent tubes. One air monitoring method was developed, tested and reported for the analysis of 2-EHN by the OSHA (Occupational Safety and Health Administration)<sup>9</sup>. However it is difficult to find a sample treatment method for 2-EHN from an aqueous sample; and, because of the quite low concentration in most environmental samples, a special concentration step using MSPD with Amberlite XAD-4 resin and Soxhlet extraction with ethyl acetate was established before GC-MS analysis. XAD-4 resin is composed of polystyrene-divinylbenzene<sup>10,11</sup> (PS/DVB; polystyrene as monomer and divinylbenzene as crosslinker) and is a well-known polymer base for ion-exchange resin and chromatography. Sun<sup>12</sup> mentioned in a thesis as "Since PS/DVB resins are stable in the pH range of 0-14, they find increasing applications in the separation of low molecular weight compounds, peptides, and proteins by means of reversed phase chromatography. For polystyrene compatible solutes, the partitioning (adsorption) process involves not only the surface of polystyrene packing material, but also the whole bulk mass."

XAD-4 (CAS No. 37380-42-0) is poly aromatic non-ionic cross-linked hydrophobic polymer used to remove small hydrophobic compounds, surfactants; widely used in pharmaceutical manufacturing; used to remove chlorinated organics, pesticides, etc. Paik etc. introduced XAD-4 Resin for the extraction of the phenolic pollutants from water sample<sup>13</sup>. And recently, Ma etc.<sup>14</sup> introduced "a method of soxhlet-assisted (SA) MSPD to extract flavonoids from rape (*Brassica campestris*) bee pollen". They showed silica gel mixed with solid sample for MSPD (for 60 min extraction time) with ethanol followed by 1 hour extraction time by modified soxhlet extractor similar to Paik's method<sup>13</sup>. But MSPD with XAD-4 for the quantitative analysis of 2-EHN from aqueous samples has not been reported yet. In this study,

a core sample preparation step of MSPD using Amberlite XAD-4 resin and mass spectral ion selection for the quantization of 2-EHN was optimized and applied for the analysis of 2-EHN in some surface water samples from South Korea.

## 2. MATERIALS AND METHODS:

### 2.1. Reagent and Material:

HPLC grade water, methanol, ethanol and ethyl acetate were provided from Fisher Scientific (Fair Lawn, NJ, USA). Whatman qualitative filter paper Grade 2, anhydrous sodium sulfate and other chemicals were purchased from Sigma-Aldrich, (St. Louis, MO, USA). The adsorbent for MSPD, Amberlite XAD-4 resin (20~60 mesh size) was purchased from Supelco (Bellefonte, PA, USA). XAD-4 resin was stored in HPLC grade methanol (in 4°C refrigerator). It was rinsed with HPLC grade water before using. A modified Soxhlet device as described by Paik et al.<sup>7</sup> was used. The bottom of the Soxhlet extractor without a siphon arm was pierced to allow the condensed solvent stream to go directly to the round-bottom flask. A cone-shaped glass thimble (filled with 1g of XAD-4 resin secured by cleaned glass wool at its tapered end) with a small hole at the end of the cone was positioned inside of the modified Soxhlet extractor. As a result, drops of condensed ethyl acetate (containing extracted 2-EHN from the XAD-4 resin) dropped continuously into the round flask, the heated reservoir containing HPLC grade ethyl acetate. Ethyl acetate was evaporated under a vacuum by Buchi R-144 Rotary Evaporator (New Castle, DE, USA). GC grade 2-EHN (higher than 99.0% purity) was purchased from Sigma-Aldrich (St. Louis, MO, USA). For the preparation of a 2-EHN stock solution, 0.5g of 2-EHN was carefully measured in a 4mL glass vial followed by dissolution with 1.5mL HPLC grade ethanol, and the resulting solution moved to a 1L volumetric flask.

The 4mL vial (measured 0.5g of 2-EHN) was washed with HPLC grade water three times. Then the water from the washing was moved to a volumetric flask containing 2-EHN and the flask filled to the 1L line with HPLC grade water (2-EHN concentration was 500mg/L). This was kept at 4°C in a refrigerator and used within 1 month. The working solution of 2-EHN was prepared by step by step dilution of the stock solution with HPLC grade water. The surface water samples were collected in turbid river water near to a GCC quarry and in clear river water far from the GCC quarry. The collected surface water samples were stored in refrigerator (4 °C).

### 2.2. Standard and Sample Preparation Method:

A recovery test with 20µg/L of 2-EHN in distilled water was performed with the modified sample treatment method of Paik et al.<sup>7</sup>. Five hundred mL of the 2-EHN

spiked sample was mixed with 1 g of activated Amberlite XAD-4 resin followed by stirring from 6 hours to 24 hours at room temperature (about 17°C). The water sample with XAD-4 resin was filtered with Whatman 2 filter paper by gravity. The recovered XAD-4 adsorbents were extracted with 80mL of HPLC grade ethyl acetate by the modified Soxhlet extractor (3 hours at 90°C). The extracted ethyl acetate was concentrated up to 1mL with a vacuum evaporator. The aliquot was analyzed by GC-MS after dehydration with anhydrous sodium sulfate followed by filtration with a 0.45µm PTFE syringe filter (Acrodisc, Pall Corporation, Port Washington, NY, USA). For the analysis of surface water samples, the supernatant acquired after centrifugation at 5000g was used as a sample. For the preparation of a standard curve, 10mg/L of 2-EHN in HPLC grade water was prepared and diluted to 5, 1, and 0.5mg/L, consecutively.

### 2.3. Instrumental Analysis:

Leco Pegasus IV (Saint Joseph, MI, USA) GC×GC-time of flight mass spectrometer (MS) was used with HP-5MS (28m × 0.25mm I.D., 0.25µm film thickness). The oven temperature was hold at 40°C for 2min and , followed by ramping with 10°C/min until 120°C and ramped with the rate 20°C/min until 270°C (hold for 5min). The mobile phase was helium (1mL/min) and the 2µL of the sample was injected by split mode (20:1). GC Injector and MS transfer line temperatures were 260°C and 240°C, respectively. Ion source temperature was set to 240°C with electron impact mode (70 eV). Mass range was m/z 45 to 400 with an acquisition rate of 20 spectra per second. The mass spectral ions of m/z 57, 70, 83, 98, and 112 were selected for the quantitative analysis. The recovery result of 2-EHN (20µg/L) was compared for each selected ion. The limit of quantization (LOQ) of 2-EHN was measured by gradual decreasing of the fortified amount (10, 5, 1, 0.5, and 0.1µg/L) of 2-EHN in 500mL distilled water. All analyses were performed in triplicate.

## 3. RESULTS AND DISCUSSION:

The quantitative analysis of 2-EHN was performed using a calibration curve with four different concentrations (0.5, 1.0, 5.0, and 10mg/L). For part per billion (ppb) level detection of 2-EHN, optimum surface water sample volume was 500mL according to preliminary experiments. The dilution factor for calculating 2-EHN concentration in the surface water sample was 500, which was derived from the surface water volume of 500mL and the final aliquot of 1mL. Therefore the four points of 0.5, 1.0, 5.0, and 10mg/L in the calibration curve were converted to 0.001, 0.002, 0.01, and 0.02mg/L. Intraday and interday precisions were 11% and 14%, respectively. The linearity of 2-EHN calibration curve was 0.994 of correlation coefficient (by

m/z 83) which is satisfactory for the quantitative analysis (Figure 1).

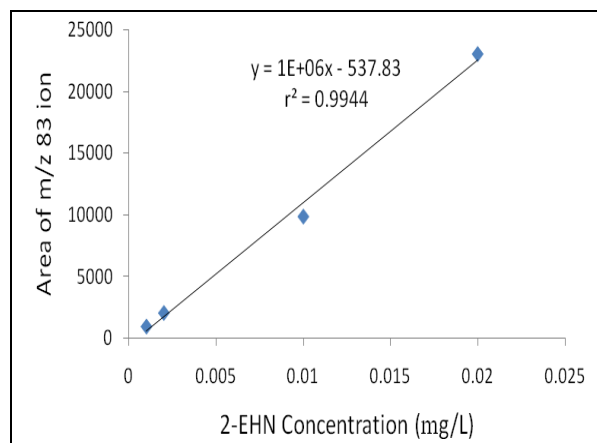


Figure 1. Standard Curve of 2-EHN in 500mL of water sample

The concentration of 2-EHN in surface water was at the ppb level, which is too low for detection without a dynamic concentration step; therefore, an effective and precise concentration step, MSPD with Amberlite XAD-4 resin, was adapted with a modified Soxhlet extraction for the 2-EHN sample preparation. An amount of XAD-4 resin of 1~2g was used to fit the internal volume (after swelling) of the glass thimble under the tapered end. The solution containing the 2-EHN was stirred at about 600 rpm during the extraction using a magnetic stirrer. Ion peak intensity of 2-EHN was compared in accordance with the XAD-4 resin (1g) at stirring times of 6, 10, 14, 18, and 24 hours. The intensity increased gradually until 14 hours with 1g of XAD-4 resin, but it then fell sharply to record 24% of the highest ion peak intensity at 14 hours with 1g of XAD-4. This indicates that 14 hours may be the maximum time period for adsorption of 2-EHN by the XAD-4 resin (1g). In comparison with the result using 1 g of XAD-4, MSPD with 14 hours stirring performed with 2g of XAD-4 showed 69% of the ion peak intensity. Moreover the precision of ion peak intensity (RSD%; Relative Standard Deviation %) was 34% for MSPD with 2g of XAD-4, which is almost four times higher than the RSD% of 9% with 1 g of XAD-4. Irregular adsorption of 2-EHN onto the oversized surface area of 2g of XAD-4 may cause the high RSD% of the 2-EHN ion peak intensity.

Mass spectrum of 2-EHN and the selected ion chromatogram of a surface water sample are shown in Figure 3. Even though the m/z 57 ion intensity is highest, the peak tailing of the m/z 57 ion dragged long, which indicates that the ion might not be of sufficient purity to represent the real intensity of the 2-EHN peak. Therefore the recovery test was performed (for the specificity evaluation<sup>15</sup>) with 500mL of distilled water fortified with 10µg of 2-EHN.

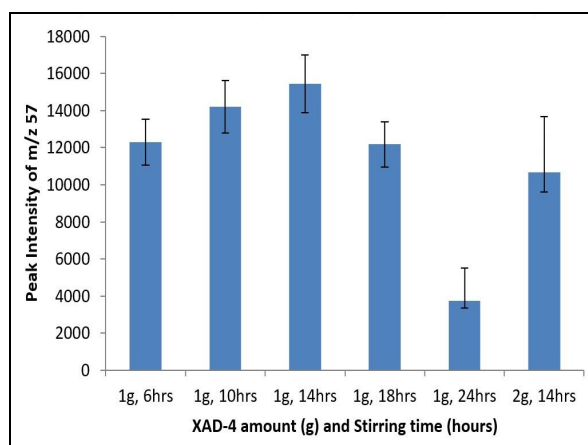


Figure 2. Recovery comparison of 2-EHN according to MSPD condition (XAD-4 resin amount and stirring time)

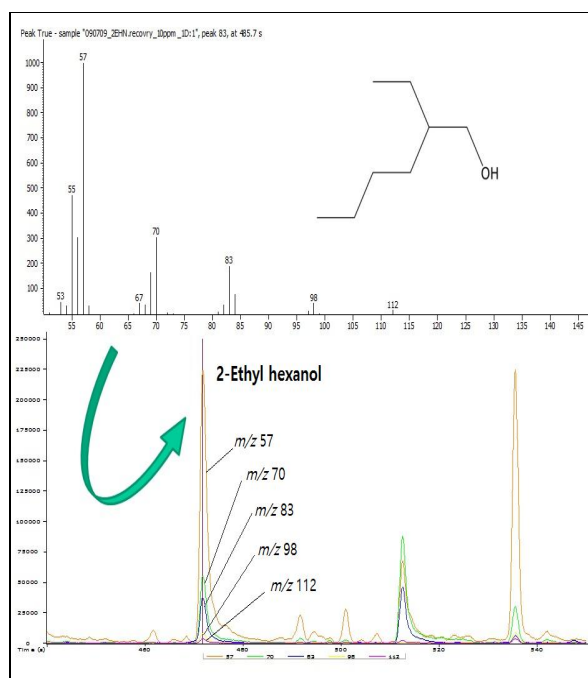


Figure 3. EI mass spectrum of 2-EHN and selected ion chromatogram of a surface water sample

The recovered 2-EHN (20µg/L) was measured by each selected mass spectral ion (m/z) (Figure 4). The recovery range of 2-EHN was from 119% to 145%. The average recovery by m/z 57, 70, 83, 98, and 112 was 131%, 138%, 126%, 126%, and 132%, respectively. The lowest RSD% was observed as 5% in the 126% recovery acquired by quantitation using m/z 83, while the RSD% of the others was 6% ~ 9%. The ions m/z 83 and 98 gave the most accurate results with 126% each, which are the values closest to 100%. This indicates that they may be the most desirable mass spectral ions for the quantitation of 2-EHN because of the lack of background noise. As mentioned above, the m/z 57 ion may cause overestimation of the 2-EHN in the water sample because of

the additional intensity caused by high background noise. LOQ of 2-EHN was evaluated as 0.5 µg/L with a signal to noise (S/N) value over 10. The lower level of 2-EHN, 0.1 µg/L, was detected as a trace level that was not precise enough for the quantitation analysis. The established sample preparation method coupled with GC-MS quantitative analysis with m/z 83 was applied to the surface water samples. The concentration of 2-EHN sampled far from and close to the GCC quarry was 3.0 and 6.8 µg/L, respectively. The surface water close to the GCC quarry with 6.8 µg/L of 2-EHN appeared more turbid than the water far from the GCC quarry.

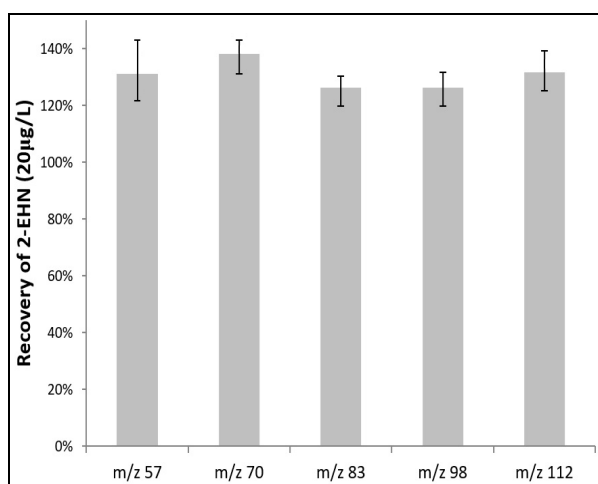


Figure 4. Recovery comparison of 2-EHN spiked to distilled water (20 µg/mL) quantitated by each selected mass spectral ion.

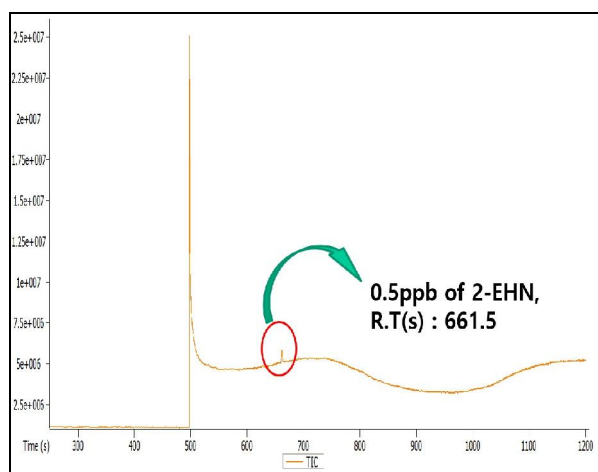


Figure 5. LOQ 0.5 µg/L of 2-EHN with S/N ratio over 10, evaluated by spiking of 2-EHN to distilled water

#### 4. CONCLUSION:

In this paper, matrix dispersive XAD-4 extraction and modified Soxhlet extraction coupled with GC-MS quantitation by a selected ion was established for measurement of 2-EHN in surface water samples. The best extraction condition for 2-EHN in this study was

found to be stirring for 14 hours with 1g of XAD-4 in a 500mL water sample. Soxhlet extraction worked well with 3 hours continuous extraction (at 90°C) with ethyl acetate by a modified Soxhlet extractor (without a siphon arm and with a glass thimble allowing condensed solvent to go through the solvent reservoir via a solid sorbent). The LOQ of 2-EHN was 0.5mg/L. The mass ions m/z 83 and 90 were the most reliable ions for the GC-MS quantitative analysis of 2-EHN in water. The concentration of 2-EHN in surface water sampled far from and close to a GCC quarry was 3.0 and 6.8 µg/L, respectively. Matrix dispersive SPE with a small amount of non-polar polystyrene resin (XAD-4) was a very reliable and precise sample preparation method for the dynamic concentration of 2-EHN in water at the ppb level.

#### 5. ACKNOWLEDGMENT:

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